Aggregate Stagnation and Sectoral Divergence: Total Factor Productivity in Colombia, 2005–2024

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Abstract

Colombia's near-zero aggregate total factor productivity (TFP) growth between 2005 and 2024 conceals economic activities moving in different directions. We reconstruct the long-run evolution of TFP for the total economy and nine activities using the annual production-account results published by the national statistical office (DANE). We chain the 19 annual log changes from 2006 to 2024, recover the annual aggregation weights implicit in the publication, and decompose aggregate TFP into exact activity contributions. The aggregate TFP index fell from 100 to 98.5, equivalent to −0.081 percentage points per year. Agricultural TFP increased by 27.6 percent, while mining TFP fell by 35.1 percent. Four activities contributed 0.361 percentage points per year to aggregate TFP, but five activities subtracted 0.441 points. Thus, aggregate stagnation was the net result of offsetting positive and negative sectoral contributions. An accounting counterfactual that sets annual TFP growth to zero in mining, construction, finance and real estate, and manufacturing raises production-account output growth from 3.15 to 3.59 percent per year. The exercise measures an accounting margin, not a causal policy effect or a counterfactual path for GDP.

Keywords: total factor productivity; growth accounting; KLEMS; sectoral productivity; Colombia.
JEL codes: O47; O54; E23; C43.

1 Introduction

Colombia's near-zero aggregate total factor productivity (TFP) growth is not evidence that every economic activity stagnated. The same aggregate result can reflect either similarly weak outcomes across activities or large positive and negative sectoral contributions that offset one another. The distinction matters because the diagnosis changes: a common slowdown calls for examining economy-wide constraints, whereas offsetting contributions require identifying what happened within the activities on each side of the balance.

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This paper asks whether Colombia's aggregate TFP stagnation between 2005 and 2024 was a common sectoral outcome or the net result of divergent activity paths. We use the production-account results published by the Departamento Administrativo Nacional de Estadística (DANE) for the total economy and nine broad economic activities. The DANE framework follows the KLEMS growth-accounting approach: output growth is decomposed into the contributions of labor, capital, energy, materials, services, and a TFP residual (Departamento Administrativo Nacional de Estadística, 2022; OECD, 2001; LA-KLEMS, 2021). The production approach provides a comparable activity history because intermediate inputs enter explicitly.

The paper makes three contributions. First, it converts DANE's annual TFP changes into chained indices for each activity and the total economy. This permits a direct comparison between the levels of 2005 and 2024. Second, it recovers the annual aggregation weights implicit in the published tables and uses them to obtain exact activity contributions to aggregate TFP. The recovery reproduces all 13 published growth-accounting components for the total economy to numerical precision. Third, it uses the same accounting identity to construct a transparent counterfactual. Annual TFP growth is set to zero in the four activities with the weakest long-run performance, while all other input contributions and annual weights remain fixed.

The main result is that aggregate stagnation was real, but it was not a common sectoral pattern. The chained index for the total economy fell by 1.5 percent. Agricultural TFP increased by 27.6 percent, whereas mining TFP fell by 35.1 percent. Social services made the largest positive contribution to aggregate TFP, at 0.148 percentage points per year. Finance and real estate made the largest negative contribution, at -0.137 points, followed by mining at -0.133 points. Manufacturing and construction each subtracted about 0.081 points per year, despite very different sectoral TFP paths and average weights.

The accounting counterfactual illustrates the aggregate importance of the negative contributions. Production-account output grew by 3.15 percent per year. If annual TFP growth in mining, construction, finance and real estate, and manufacturing had been zero, the same accounting identity yields growth of 3.59 percent per year. The difference is 0.43 percentage points. This is not an estimate of a feasible policy intervention. It does not model changes in factor demand, prices, input use, or resource allocation, and it does not refer to GDP growth.

The rest of the paper is organized as follows. Section 2 describes the DANE data, the KLEMS identity, the long-run chaining procedure, the activity decomposition, and the counterfactual. Section 3 presents the aggregate and sectoral results. Section 4 discusses interpretation, limitations, and policy questions. Section 5 concludes. The appendix documents the weight recovery, additional growth-accounting results, subperiod contributions, and the activity classification.

2 Data and Accounting Framework

2.1 DANE production-account data

DANE publishes annual growth-accounting results for the total economy and nine activity aggregates (Departamento Administrativo Nacional de Estadística, 2025). The source annex reports output growth; contributions from labor composition and hours; contributions from information and communication technology (ICT) and non-ICT capital; contributions from energy, materials, and services; and TFP. The activity classification follows CIIU Rev. 3/3.1 as implemented in Colombia. The appendix lists the activities included in each aggregate.

The analysis uses the 19 annual changes from 2006 through 2024. These observations connect the level in 2005 with the level in 2024. The row labeled 2005 in the DANE annex measures the change from 2004 to 2005 and is therefore not part of the 2005–2024 comparison.

For activity i and year t, the production-account identity is

$$\Delta \ln Y_{i,t} = C^L_{i,t} + C^K_{i,t} + C^E_{i,t} + C^M_{i,t} + C^S_{i,t} + \Delta \ln A_{i,t}, \tag{1}$$

where Y is output, A is TFP, and C^L, C^K, C^E, C^M, and C^S are the measured contributions of labor, capital, energy, materials, and services. Labor includes changes in hours and labor composition. Capital includes ICT and non-ICT assets. Each contribution combines input growth with its cost share; it is not the growth rate of the corresponding input. TFP is the residual between output growth and the measured input contributions.

This residual has a precise accounting meaning but not a unique structural interpretation. Under the assumptions used in growth accounting, it is commonly associated with changes in efficiency or technology (Solow, 1957; Jorgenson et al., 1987). In annual activity data, it may also capture changes in capacity utilization, reallocation, omitted inputs, and measurement error (OECD, 2001). We therefore describe the TFP paths and formulate possible mechanisms as hypotheses rather than identified causes.

2.2 Chaining annual TFP changes

Let $I_{i,t}$ denote the TFP index for activity i, normalized to $I_{i,2005} = 100$. Because the annex reports log changes in percentage points, the index evolves according to

$$I_{i,t} = I_{i,t-1} \exp\left(\frac{\Delta \ln A_{i,t}}{100}\right). \tag{2}$$

The cumulative percentage change between 2005 and 2024 is

$$G_i = 100\left[\exp\left(\frac{\sum_{t=2006}^{2024} \Delta \ln A_{i,t}}{100}\right) - 1\right], \tag{3}$$

and the annualized log change is

$$\overline{g_i} = \frac{1}{19}\sum_{t=2006}^{2024} \Delta \ln A_{i,t} = \frac{100}{19} \ln\left(\frac{I_{i,2024}}{I_{i,2005}}\right). \tag{4}$$

The cumulative percentage change and the annualized log change describe TFP within an activity. They cannot be added across activities.

2.3 Aggregate TFP and activity contributions

DANE states that aggregate productivity and its components are constructed as Törnqvist-weighted sums of activity growth rates (Departamento Administrativo Nacional de Estadística, 2022). Törnqvist aggregation belongs to the superlative index-number methods used to approximate changes in flexible production structures (Diewert, 1976). The annual aggregate TFP identity can therefore be written as

$$\Delta \ln A_t = \sum_{i=1}^{9} w_{i,t} \Delta \ln A_{i,t}, \qquad \sum_{i=1}^{9} w_{i,t} = 1, \tag{5}$$

where $w_{i,t}$ is the annual aggregation weight. The public annex does not report these weights. We recover one vector for each year by requiring the same weights to reproduce eight independent published components for the total economy and by imposing that the nine weights sum to one. The recovered weights also reproduce the five remaining columns, including aggregate TFP, even though those columns are not used to identify the system. Appendix A describes this validation.

The contribution of activity i in year t is

$$c_{i,t} = w_{i,t} \Delta \ln A_{i,t}. \tag{6}$$

Its average annual contribution over the full period is

$$\overline{c_i} = \frac{1}{19} \sum_{t=2006}^{2024} c_{i,t}. \tag{7}$$

By construction,

$$\overline{\Delta \ln A} = \sum_{i=1}^{9} \overline{c_i}. \tag{8}$$

This operation differs from applying a single set of average weights to cumulative sectoral growth. Let $\widetilde{w}_i$ denote the mean annual weight. Then

$$\overline{c_i} = \widetilde{w}_i \, \overline{g_i} + \mathrm{Cov}_t \left(w_{i,t}, \Delta \ln A_{i,t} \right). \tag{9}$$

The covariance term matters whenever activity weights and TFP growth move together. In the data, multiplying average weights by annualized sectoral TFP yields aggregate TFP of -0.088 percentage points per year. The exact year-by-year calculation yields -0.081 points; the 0.007-point difference is the sum of the covariance terms.

2.4 Accounting counterfactual

Let $\mathcal{S}$ contain mining, construction, finance and real estate, and manufacturing. These are the four activities with the lowest annualized TFP growth over 2006–2024. The counterfactual replaces their annual TFP growth with zero:

$$\Delta \ln A_t^{CF} = \Delta \ln A_t - \sum_{i \in \mathcal{S}} c_{i,t}. \tag{10}$$

All other input contributions and the recovered annual weights remain fixed. The production-account identity then implies

$$\Delta \ln Y_t^{CF} = \Delta \ln Y_t - \sum_{i \in \mathcal{S}} c_{i,t}. \tag{11}$$

The exercise is deliberately mechanical. Activity selection uses the full sample, the counterfactual does not specify an intervention, and it does not allow labor, capital, intermediate inputs, prices, or weights to respond. Its purpose is to measure the accounting importance of the four negative long-run contributions.

3 Results

3.1 Aggregate TFP ended below its 2005 level

Figure 1 reports the chained TFP index for the total economy. The index increased from 100 in 2005 to 102.2 in 2006, then declined and remained below 100 from 2009 onward. It reached 98.5 in 2024. The cumulative decline was 1.5 percent, equivalent to an annualized log change of −0.081 percentage points.

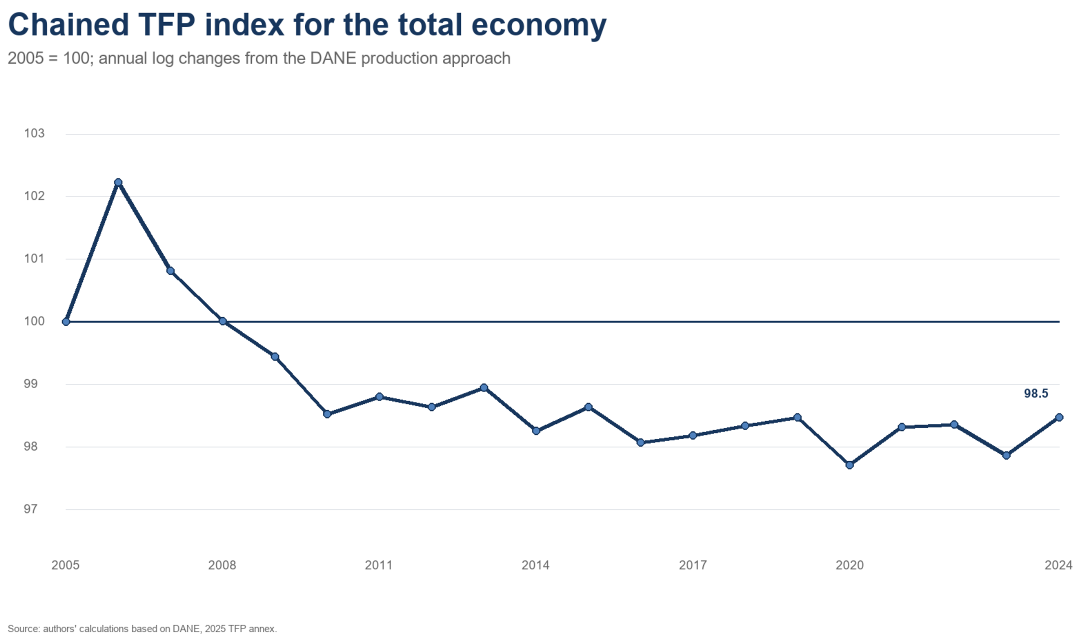

Figure 1: Chained TFP index for the total economy, 2005–2024

Most of the aggregate decline occurred before 2011. Aggregate TFP averaged −0.296 percentage points per year in 2006–2010. The subsequent averages were close to zero: 0.022 points in 2011–2015, −0.040 in 2016–2019, and −0.001 in 2020–2024. The early deterioration stopped, but aggregate TFP did not enter a sustained growth path.

3.2 Aggregate stagnation concealed sectoral paths moving in opposite directions

Figure 2 displays the 19 annual TFP changes that determine the long-run results. The activities are ranked from the lowest to the highest cumulative TFP growth, with the total economy shown in the final row. Mining recorded negative TFP growth in 17 of 19 years and manufacturing in 15. Agriculture recorded positive TFP growth in 15 years; trade and social services did so in 14.

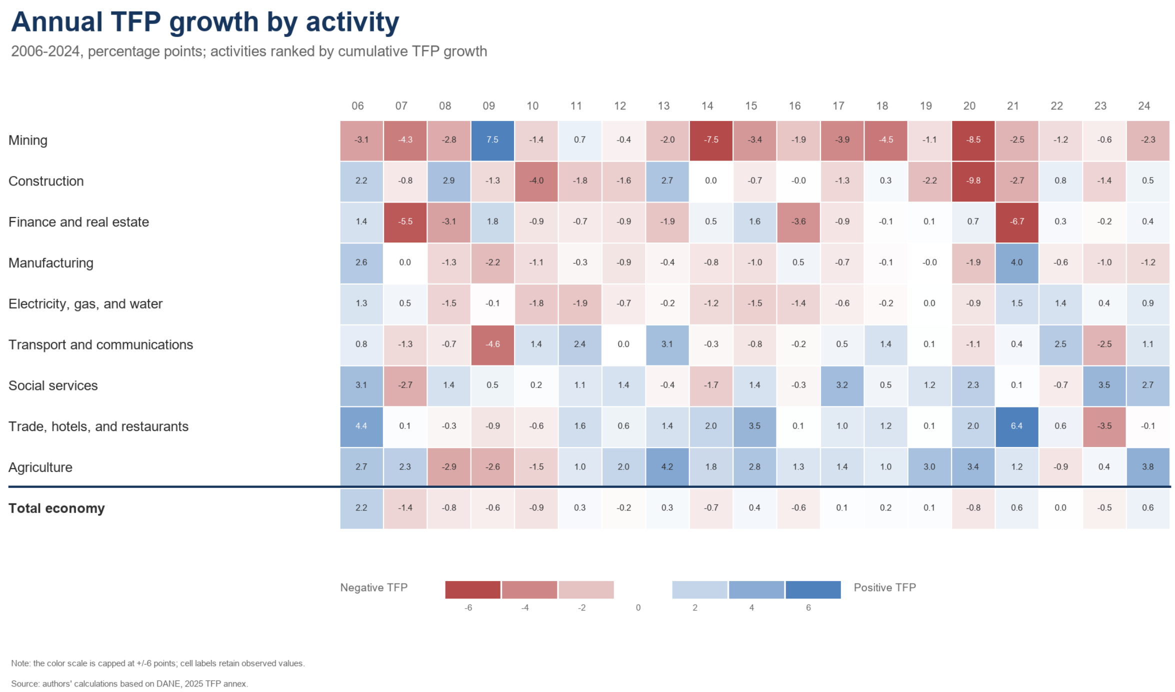

Figure 2: Annual TFP growth for the total economy and nine activities, 2006–2024

The annual observations also show why long-run averages should not be interpreted as smooth trends. Construction TFP fell by 9.8 percentage points in 2020. Trade TFP increased by 6.4 points in 2021, while finance and real estate fell by 6.7 points. These changes affect the full-period averages, but the sign frequencies show that the strongest long-run results are not produced by a single year alone.

Figure 3 reports the cumulative and annualized measures. Agriculture had the highest cumulative TFP growth, 27.6 percent, equivalent to 1.28 percentage points per year. Trade increased by 21.5 percent and social services by 18.4 percent. Mining had the largest decline, 35.1 percent, equivalent to -2.28 points per year. Construction and finance and real estate each lost about 16 percent, while manufacturing fell by 6.5 percent.

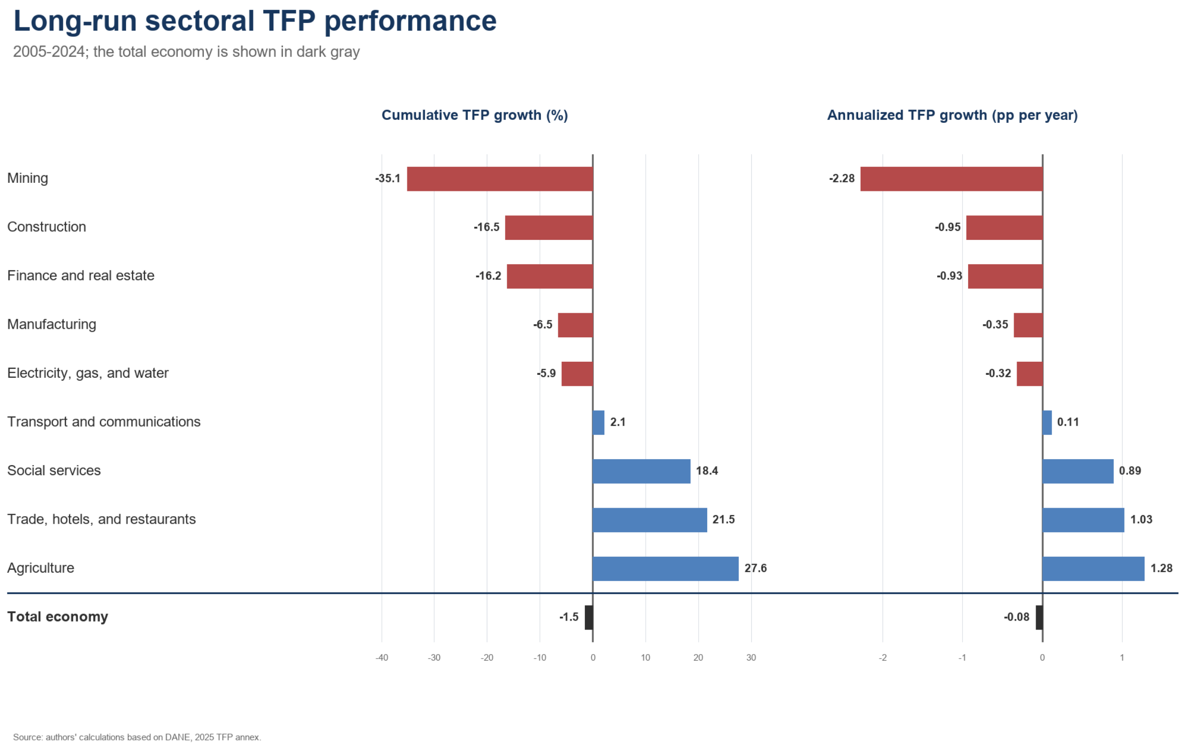

Figure 3: Cumulative and annualized TFP growth by activity, 2005–2024

3.3 Positive and negative sectoral contributions nearly offset

Sectoral TFP growth does not determine an activity's aggregate contribution on its own. The contribution also depends on its annual aggregation weight. Figure 4 and Table 1 report both dimensions. Four activities contributed 6.852 cumulative log points, or 0.361 percentage points per year. Five activities subtracted 8.388 cumulative log points, or 0.441 points per year. The net contribution was -1.535 log points, equivalent to the aggregate annual average of -0.081 points.

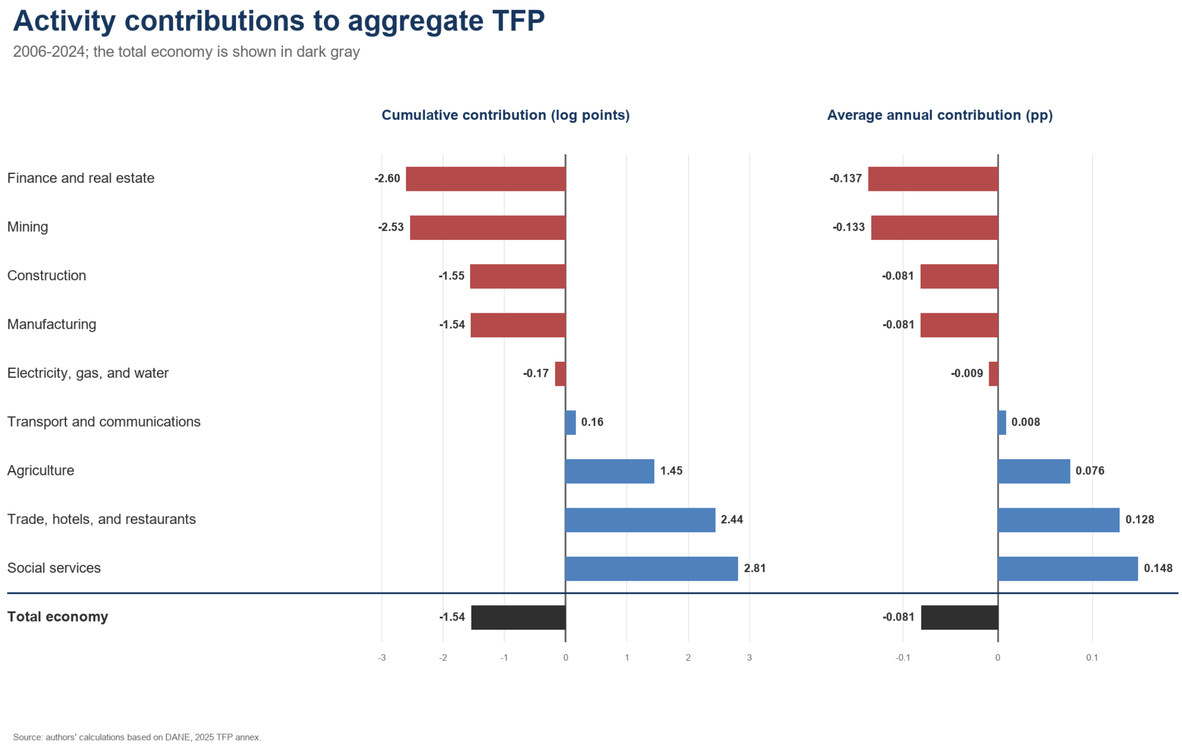

Figure 4: Activity contributions to aggregate TFP, 2006–2024

Table 1: Sectoral TFP performance and contribution to the total economy, 2005–2024

Activity	Average weight (%)	Cumulative TFP growth (%)	Annualized TFP (pp per year)	Contribution (pp per year)
Social services	16.08	18.44	0.89	0.148
Trade, hotels, and restaurants	13.00	21.54	1.03	0.128
Agriculture	6.07	27.59	1.28	0.076
Transport and communications	8.62	2.15	0.11	0.008
Electricity, gas, and water	3.43	-5.85	-0.32	-0.009
Manufacturing	23.26	-6.51	-0.35	-0.081
Construction	8.71	-16.53	-0.95	-0.081
Mining	6.13	-35.13	-2.28	-0.133
Finance and real estate	14.70	-16.16	-0.93	-0.137
Total economy	**100.00**	**-1.52**	**-0.08**	**-0.081**

Notes: Cumulative growth compares the 2024 chained index with 2005. Annualized TFP is the mean log change. Sectoral cumulative and annualized TFP describe performance within each activity and do not add across activities. The contribution multiplies annual activity TFP by its annual weight before averaging; the nine contributions add to aggregate TFP. Figures may not add because of rounding.
Source: Authors' calculations based on DANE, 2025 TFP annex.

Social services made the largest positive contribution, at 0.148 percentage points per year. Its average weight was 16.1 percent and its own TFP increased by 0.89 points per year. Trade contributed 0.128 points. Agriculture had the strongest sectoral TFP growth but a lower average weight of 6.1 percent, so its contribution was 0.076 points. Transport contributed 0.008 points.

Finance and real estate made the largest negative contribution, at −0.137 percentage points per year. Its own TFP fell by 0.93 points and its average weight was 14.7 percent. Mining subtracted 0.133 points. Its average weight was only 6.1 percent, but its own TFP decline was much larger at 2.28 points per year.

Manufacturing and construction each subtracted about 0.081 percentage points per year for different reasons. Manufacturing had the largest average weight, 23.3 percent, but its own TFP fell by 0.35 points per year. Construction had an average weight of 8.7 percent and a larger sectoral decline of 0.95 points. These cases show why rankings based only on sectoral TFP growth do not identify the activities with the largest aggregate effect.

3.4 A zero-TFP counterfactual raises production-account output growth

Figure 5 shows the accounting counterfactual. Production-account output grew by 3.15 percent per year between 2006 and 2024. Setting annual TFP growth to zero in the four weakest activities raises the average to 3.59 percent. The difference is 0.433 percentage points per year.

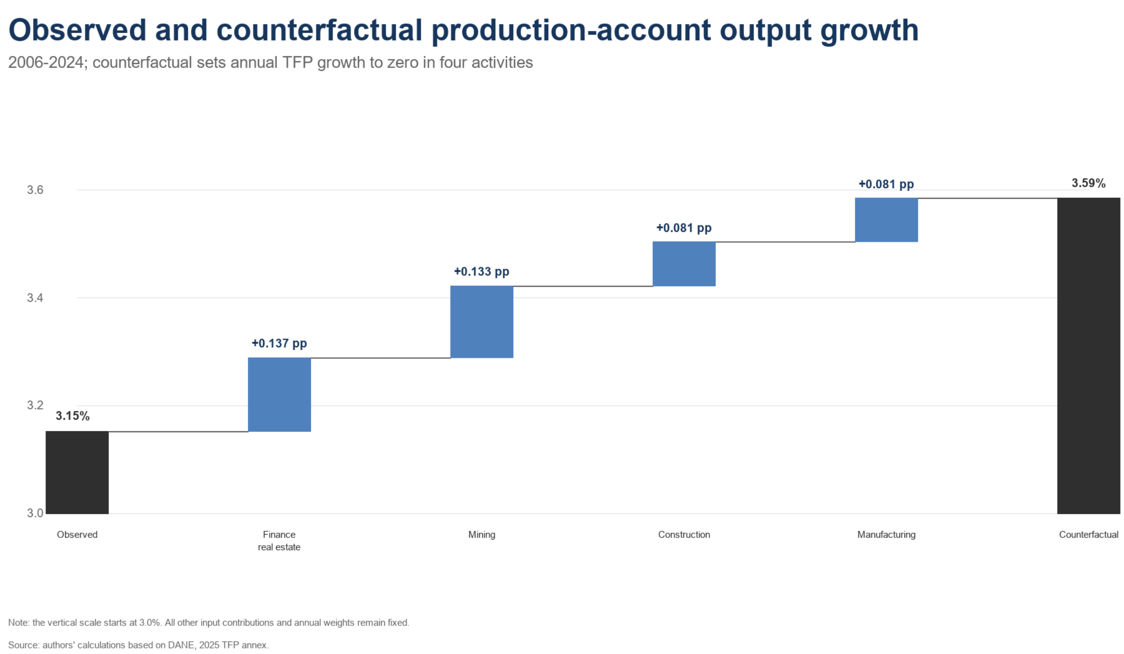

Figure 5: Observed and counterfactual production-account output growth, 2006–2024

Finance and real estate and mining account for 0.270 of the 0.433 additional points. Setting finance and real estate TFP to zero adds 0.137 points per year, and doing the same for mining adds 0.133. Construction and manufacturing each add 0.081 points. These changes equal the negative of their average TFP contributions because the exercise holds all other components fixed.

The counterfactual raises output growth in 17 of the 19 years. It lowers growth in 2006 and 2009 because the four selected activities made a positive joint TFP contribution in those years. Compounding the annual differences implies a production-account output level in 2024 that is 8.6 percent above the observed level.

This result should not be interpreted as a policy estimate. It shows the amount of observed growth associated with the negative TFP contributions under the KLEMS identity. It does not show that zero TFP growth was feasible, which intervention could have achieved it, or how factor use and relative prices would have responded.

4 Discussion

Aggregate stagnation is real, but not a common sectoral outcome. Colombia's production-account TFP was nearly flat over 2005–2024 because positive contributions from social services, trade, agriculture, and transport were almost fully offset by finance and real estate, mining, construction, manufacturing, and electricity. The relevant empirical question is therefore not only why aggregate TFP stagnated. It is why productivity gains persisted in some activities while negative contributions remained large in others.

The results do not identify the mechanisms behind each activity path. Mining illustrates the limitation. Its output grew by 1.70 percent per year, but the measured input contributions exceeded that growth and produced TFP of -2.28 points. This pattern may reflect resource depletion, the composition of extraction, investment cycles, capacity utilization, regulation, or measurement. The data used here cannot distinguish among these explanations.

Finance and real estate present a different problem. The combined aggregate covers financial intermediation, real estate, and business and rental activities. Its negative contribution may therefore combine distinct subactivities. Output and price measurement are also difficult in financial and real-estate services. A useful extension would reconstruct the contribution at a finer level before assigning a common mechanism to the aggregate.

Manufacturing and construction show why both the sectoral change and the aggregation weight matter. Manufacturing had a moderate TFP decline but a large average weight. Construction had a sharper decline and a smaller weight. Their average aggregate contributions were almost identical. A policy diagnosis based only on the largest sectoral decline would miss manufacturing's aggregate relevance.

The counterfactual provides a scale for these negative contributions, not a policy target. Raising sectoral TFP requires changes in technology, organization, regulation, factor allocation, capacity use, or measurement, depending on the activity. A credible policy exercise must replace the zero-TFP benchmark with attainable sector-specific references and allow labor, capital, intermediate inputs, prices, and activity weights to respond.

The next empirical step should therefore be sector-specific. For manufacturing, the evidence should test the roles of technology adoption, management practices, exporting, and firm reallocation. For construction, it should distinguish project-cycle and capacity-utilization effects from persistent weaknesses in project management and production methods. For mining, it should separate resource and composition effects from investment and regulatory conditions. For finance and real estate, the first requirement is a more detailed statistical decomposition.

The analysis also has two general limitations. First, the period begins in 2005 because the sectoral production-account history in the annex is reported as annual changes rather than a longer published index. Second, the recovered weights are implicit in the DANE aggregation. Their exact reconciliation supports the contribution calculation, but the public annex does not expose the underlying nominal series used to construct them. Replication therefore validates the accounting system, not every stage of DANE's source-data construction.

5 Conclusion

Aggregate TFP stagnation in Colombia between 2005 and 2024 was not a common outcome across economic activities. The chained index for the total economy fell from 100 to 98.5, a cumulative decline of 1.5 percent and an annualized log change of -0.081 percentage points. Agricultural TFP increased by 27.6 percent, while mining TFP fell by 35.1 percent. The aggregate result therefore conceals activities moving in opposite directions.

The near-zero total was the balance of contributions with opposite signs. Social services, trade, agriculture, and transport contributed 0.361 percentage points per year. Finance and real estate, mining, construction, manufacturing, and electricity subtracted 0.441 points. Finance and real estate and mining made the largest negative contributions, while social services made the largest positive contribution.

The year-by-year aggregation is central to this conclusion. Cumulative sectoral TFP growth does not add across activities, and average weights applied to long-run sectoral growth do not reproduce the exact total when weights change over time. Recovering the annual weights and averaging annual contributions preserves the published aggregate identity.

The accounting counterfactual shows the scale of the negative contributions. If annual TFP growth in the four weakest activities had been zero, production-account output growth would have been 3.59 percent per year rather than 3.15 percent. The 0.43-point difference is economically meaningful, but it is not a causal estimate or a forecast of GDP. The result identifies where a more detailed diagnosis may have the highest aggregate payoff. It does not identify the policies that would produce those gains.

A Annual Aggregation Weights

For each year, we recover nine activity weights. The system uses the activity and total-economy rows for eight independent components: output, labor composition, hours, ICT capital, non-ICT capital, energy, materials, and services. A ninth equation requires the weights to sum to one:

$$\begin{bmatrix} Z^1_{1,t} & \cdots & Z^1_{9,t} \\ \vdots & \ddots & \vdots \\ Z^8_{1,t} & \cdots & Z^8_{9,t} \\ 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} w_{1,t} \\ \vdots \\ w_{9,t} \end{bmatrix} = \begin{bmatrix} Z^1_t \\ \vdots \\ Z^8_t \\ 1 \end{bmatrix}, \tag{12}$$

where $Z^k_{i,t}$ is component k for activity i, and Z^k_t is the corresponding published total. The recovered weights range from 3.0 to 26.8 percent across the 19 years. Applying them to all 13 columns, including the five columns not used as independent equations, produces a maximum reconciliation error of 5.3×10^{-15}. The procedure identifies the weights implicit in the published aggregation; it does not reconstruct DANE's underlying nominal value series.

B Average Growth-Accounting Decomposition

Table 2: Average annual decomposition of output growth by activity, 2006–2024

Activity	Output	Labor	Capital	Energy	Materials	Services	TFP
Total economy	**3.15**	**0.87**	**0.93**	**0.20**	**0.49**	**0.73**	**-0.08**
Agriculture	2.62	-0.26	0.62	0.07	0.73	0.17	1.28
Mining	1.70	0.74	2.99	0.20	0.01	0.05	-2.28
Manufacturing	1.92	0.19	0.77	0.27	0.90	0.14	-0.35
Electricity, gas, and water	2.67	0.22	1.25	1.33	0.06	0.13	-0.32
Construction	1.94	0.69	0.84	0.03	1.24	0.09	-0.95
Trade, hotels, and restaurants	3.71	0.84	0.16	0.10	0.47	1.11	1.03
Transport and communications	4.05	1.20	0.70	0.59	0.11	1.33	0.11
Finance and real estate	3.90	1.70	2.04	0.01	0.05	1.03	-0.93
Social services	4.59	1.68	0.07	0.04	0.31	1.60	0.89

Notes: Values are average annual percentage-point contributions, except output, which is the average annual log growth rate. The 19 observations from 2006 to 2024 connect the levels in 2005 and 2024. The total-economy row is the DANE published aggregate, not a simple sum or average of the activity rows. Components may not add exactly because of rounding.

Source: Authors' calculations based on DANE, 2025 TFP annex.

C Contributions by Subperiod

Table 3: Activity contributions to aggregate TFP by subperiod

Activity	2006–10	2011–15	2016–19	2020–24	2006–24
Agriculture	-0.019	0.122	0.094	0.110	0.076
Mining	-0.037	-0.193	-0.155	-0.153	-0.133
Manufacturing	-0.103	-0.157	-0.021	-0.031	-0.081
Electricity, gas, and water	-0.010	-0.035	-0.018	0.026	-0.009
Construction	-0.023	-0.022	-0.078	-0.202	-0.081
Trade, hotels, and restaurants	0.069	0.216	0.077	0.140	0.128
Transport and communications	-0.079	0.076	0.038	0.005	0.008
Finance and real estate	-0.173	-0.038	-0.169	-0.173	-0.137
Social services	0.078	0.054	0.192	0.276	0.148
Total economy	**-0.296**	**0.022**	**-0.040**	**-0.001**	**-0.081**

Notes: Average annual percentage-point contributions. Each activity entry is the subperiod mean of annual TFP multiplied by the annual aggregation weight. Subperiods have different lengths.
Source: Authors' calculations based on DANE, 2025 TFP annex.

D Activity Coverage

Table 4: Correspondence between KLEMS aggregates and CIIU activities

Activity aggregate	Included CIIU activities
Agriculture	**Section A:** agriculture, livestock, hunting, and forestry. **Section B:** fishing.
Mining	**Section C:** mining and quarrying.
Manufacturing	**Section D:** manufacturing, including food and beverages; tobacco; textiles and apparel; leather and footwear; wood; paper and printing; petroleum refining; chemicals; rubber and plastics; non-metallic mineral products; metals; machinery and equipment; electrical, electronic, and transport equipment; and other manufacturing.
Electricity, gas, and water	**Section E:** electricity, gas, and water supply.
Construction	**Section F:** construction.
Trade, hotels, and restaurants	**Section G:** wholesale and retail trade; repair of motor vehicles, motorcycles, personal effects, and household goods. **Section H:** hotels and restaurants.
Transport and communications	**Section I:** transport, storage, and communications.

Activity aggregate	Included CIIU activities
Finance and real estate	**Section J:** financial intermediation. **Section K:** real estate, renting, and business activities.
Social services	**Section L:** public administration and defense; compulsory social security. **Section M:** education. **Section N:** health and social work. **Section O:** other community, social, and personal service activities. **Section P:** private households with employed persons and undifferentiated production activities of private households. **Section Q:** extraterritorial organizations and bodies.

Notes: The table documents the correspondence between the nine aggregates in the DANE annex and the CIIU Rev. 3/3.1 sections adapted for Colombia. Labels may vary across revisions without changing the KLEMS aggregate used by DANE.

Source: DANE, CIIU Rev. 3 and Rev. 3.1 A.C.; 2025 TFP annex.

Data and Code Availability

The DANE source annex, processed data, replication programs, figures, and table sources are available at:

https://github.com/oliverpardo1979/informe_ptf.

The original DANE annex is redistributed in the repository for reproducibility. Users should consult DANE for the official release and any subsequent revisions.

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